

HR ANALYTICS – EMPLOYEE ATTRITION

Employee Attrition Analysis & Workforce Insights

Project Report

Tools: Python | Pandas | PostgreSQL | SQL | Power BI | DAX

1. Executive Summary

This project analyses the IBM HR Analytics Employee Attrition dataset to identify patterns in employee turnover. The dataset contains **1,470 employees**, of whom **237** left the organisation. The resulting observed attrition rate is **16.12%**. The analysis combines Python exploratory analysis, PostgreSQL/SQL analysis and an interactive Power BI dashboard.

Metric	Result
Total Employees	1,470
Employees Left	237
Employees Stayed	1,233
Attrition Rate	16.12%
Average Age	36.92 years
Average Monthly Income	\$6,502.93

2. Business Problem

Employee attrition can increase recruitment costs, reduce productivity and create workforce knowledge gaps. The purpose of this project is to understand where attrition is concentrated and which employee, job and workplace characteristics show higher observed attrition rates. The findings are intended to support HR teams in prioritising retention investigations and workforce planning.

3. Objectives

- Measure overall employee attrition.
- Compare attrition across departments and job roles.
- Investigate age, gender and education patterns.
- Examine overtime, job satisfaction and work-life balance.
- Compare salary and monthly income patterns.
- Analyse employee tenure and experience.
- Translate analytical findings into business recommendations.

4. Data & Methodology

The cleaned dataset used for this report was supplied as **HR_Employee_Attrition_Cleaned.csv**. The target variable is **Attrition**, where Yes indicates that an employee left and No indicates that the employee stayed. Python was used for data inspection, grouping, KPI calculation and exploratory analysis. PostgreSQL was used for structured SQL analysis. Power BI was used to create the final interactive dashboard.

Key analytical fields include:

- Age, Gender and EducationField
- Department and JobRole
- MonthlyIncome
- OverTime
- JobSatisfaction
- WorkLifeBalance
- YearsAtCompany and TotalWorkingYears
- Attrition

5. Department Analysis

Department	Employees	Left	Attrition Rate
Sales	446	92	20.63%
Human Resources	63	12	19.05%
Research & Development	961	133	13.84%

Finding: Sales has the highest observed department attrition rate at 20.63%, while Research & Development has the lowest at 13.84%.

6. Age Analysis

Age Group	Employees	Left	Attrition Rate
18-25	123	44	35.77%
26-35	606	116	19.14%
36-45	468	43	9.19%
46-55	226	26	11.50%
56+	47	8	17.02%

Finding: The 18-25 group has the highest observed attrition rate at 35.77%. The analysis shows a generally higher attrition level among younger employees.

7. Overtime Analysis

Overtime	Employees	Left	Attrition Rate
No	1,054	110	10.44%

Overtime	Employees	Left	Attrition Rate
Yes	416	127	30.53%

Finding: Employees working overtime have an observed attrition rate of 30.53%, compared with 10.44% for employees not working overtime. This is one of the strongest descriptive differences in the dataset.

8. Job Role Analysis

Job Role	Employees	Left	Attrition Rate
Sales Representative	83	33	39.76%
Laboratory Technician	259	62	23.94%
Human Resources	52	12	23.08%
Sales Executive	326	57	17.48%
Research Scientist	292	47	16.10%
Manufacturing Director	145	10	6.90%
Healthcare Representative	131	9	6.87%
Manager	102	5	4.90%
Research Director	80	2	2.50%

Finding: Sales Representative has the highest observed attrition rate among job roles at 39.76%. Job-role comparisons should also consider the size of each group.

9. Salary & Income Analysis

Salary Category	Employees	Left	Attrition Rate
Low	749	163	21.76%
Medium	440	49	11.14%
High	281	25	8.90%

Finding: The Low salary category has the highest observed attrition rate at 21.76%, compared with 8.90% for the High category.

10. Gender Analysis

Gender	Employees	Left	Attrition Rate
Female	588	87	14.80%
Male	882	150	17.01%

Finding: The observed attrition rate is 17.01% for male employees and 14.80% for female employees. This is a descriptive comparison and does not establish causation.

11. Job Satisfaction & Work-Life Balance

Employees with Job Satisfaction score 1 have an observed attrition rate of 22.84%, while score 4 has an observed rate of 11.33%.

Employees with Work-Life Balance score 1 have an observed attrition rate of 31.25%. The pattern is not perfectly linear across all scores, so this factor should be interpreted together with other workforce characteristics.

12. Power BI Dashboard

The final dashboard presents the main HR KPIs and supporting visual analysis. It includes Total Employees, Employees Left, Attrition Rate, Average Age and Average Monthly Income, alongside department, gender, overtime, satisfaction, age, income, work-life balance and education analysis. Interactive filters are provided for Department, Job Role, Education Field and Gender.

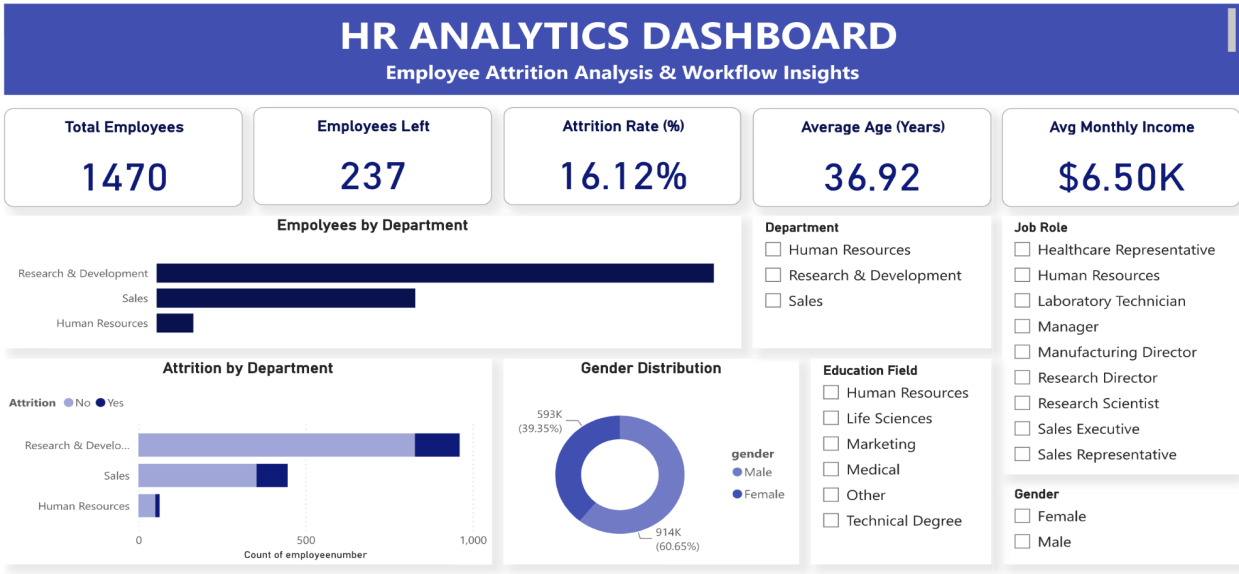




Figure 2. HR Analytics dashboard – supporting analysis page.

13. Key Business Insights

- Overall observed attrition is 16.12% across 1,470 employees.
- The 18-25 age group records the highest observed attrition rate (35.77%).
- Overtime employees have a substantially higher observed attrition rate (30.53%) than non-overtime employees (10.44%).
- Sales Representative has the highest observed attrition rate among job roles (39.76%).
- The Low salary category has the highest observed attrition rate (21.76%).
- The department with the highest observed attrition rate is Sales (20.63%).

14. Recommendations

1. Strengthen onboarding, mentoring and career-development support for early-career employees.
2. Review overtime workload, staffing levels and workload distribution.
3. Develop targeted retention plans for high-attrition job roles, particularly Sales Representatives.
4. Use regular employee feedback and engagement surveys to identify satisfaction issues.
5. Review work-life balance practices, scheduling and flexibility.
6. Investigate compensation and career progression for lower-paid employee groups.
7. Use the dashboard as a monitoring tool and investigate changes in attrition over time.

15. Limitations

- The analysis is descriptive and does not prove that any factor causes employee attrition.
- Some employee groups are relatively small, making their percentage rates less stable.
- The project does not predict individual employee attrition probability.

- Further statistical testing and predictive modelling would be needed for a production-level retention model.

16. Future Scope

- Feature engineering and statistical significance testing.
- Employee segmentation and retention-risk scoring.
- Machine-learning classification models.
- Model evaluation using precision, recall, F1-score and ROC-AUC.
- Explainable AI to understand the drivers of individual predictions.
- Automated HR reporting and dashboard refresh workflows.

17. Conclusion

This project demonstrates an end-to-end HR analytics workflow using Python, PostgreSQL/SQL and Power BI. The analysis identified meaningful differences in observed attrition across age, overtime, job role, department, salary, job satisfaction and work-life balance. With 1,470 employees and an observed attrition rate of 16.12%, the dashboard provides HR stakeholders with a concise view of workforce patterns and areas that may benefit from further investigation.

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Data Source: IBM HR Analytics Employee Attrition Dataset

Tools: Python | PostgreSQL | SQL | Power BI